Shreya Pramanik

Minneapolis, MN

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Education

University of Minnesota | **Sep 2023 – Sept 2026**
Master of Statistics | *Minneapolis, United States*
Coursework: Data Mining, Computer Vision, Regression Analysis, Advanced Probability, Bayesian Analysis

Indian Institute of Technology Kanpur | **Sep 2020 – Jun 2022**
Master of Statistics | *Kanpur, India*
Coursework: Statistical Simulation, Time Series Analysis, Design of Experiment, Applied Stochastic Process

Presidency University | **July 2017 – July 2020**
Bachelor of Statistics | *Kolkata, India*

Technical Skills

Languages & Frameworks: Python, R, SQL (MySQL, Google BigQuery), SAS, MATLAB, Excel, Power BI (DAX, Power Query), Tableau, PyTorch, TensorFlow, scikit-learn, ETL Pipelines, Salesforce

Cloud & DevOps Tools: AWS, Azure, Databricks, Docker, Snowflake, Git, Airflow, n8n

Experience

Carlson School of Management, University of Minnesota | **Feb 2026 – May 2026**
Graduate Research Assistant | *Minneapolis, MN*

- Screened 654 survey responses across three waves using long-string analysis and median completion-time checks, flagging and removing ∼5% low-quality entries to deliver a clean, analysis-ready dataset.
- Achieved **Cohen's Kappa of 0.78** (substantial agreement) on 16 qualitative interview transcripts for a longitudinal reduced work-week study, helping in policy analysis for a Minneapolis non-profit.

School of Statistics, University of Minnesota | **Sep 2023 – Jan 2026**
Graduate Teaching Assistant | *Minneapolis, MN*

- Led discussion sections and coding labs for regression and simulation courses; mentored 100+ grad students on statistical programming in R

HSBC Electronic Data Processing Pvt. Ltd | **July 2022 – July 2023**
Data Analyst, Wealth and Personal Banking | *Bengaluru, India*

- **Queried** and extracted large-scale customer contact datasets via **SQL** across 5+ APAC contact centers; built Excel-based reporting tools using **pivot tables, lookup formulas**, and automated **Power BI KPI summaries** to clean, structure, and surface trends.
- Developed and deployed an ensemble of short-term demand forecasting models using **XGBoost, LSTM, and Prophet** across **5+** APAC contact centers, achieving **80–86%** accuracy, informing real-time staffing decisions for hundreds of daily customer interactions.
- Led production-ready modeling pipeline on Malaysia customer-contact data to detect demand surges during digital-banking rollouts, informing **workforce planning decisions** to business stakeholders.

Projects

E-Commerce Analytics Platform | ***MySQL, Python, Power BI, DAX*** | **Jun 2026 – Jul 2026**

- Built an end-to-end **ETL pipeline** using Python (pandas, SQLAlchemy) to ingest **100K+ real transactions** across a 3-layer medallion architecture (staging → 3NF → star schema), normalising raw Olist e-commerce CSVs into **8 relational tables** with FK constraints enforced at the database level.
- Designed a **5-page Power BI dashboard** tracking **$13.17M in revenue** across 95K orders, identifying checkout as the highest drop-off stage (**11.56%**), organic search as the top-performing channel (**62x more efficient** than paid search), and VIP customers spending **2.3x more** than first-time buyers; supported by 6 production-grade SQL queries using window functions, recursive CTEs, and NTILE-based RFM segmentation.
- Engineered an **A/B testing framework** with deterministic variant assignment via MD5 hashing in Python and validated a statistically significant **31% conversion lift** (11.71% → 15.36%, $z = 9.24 > 1.96$) through a **SQL-based z-test**, while implementing **CI/CD via GitHub Actions** with automated SQL linting and pytest on every push.

Financial Shock Effects on Hospital Operations | ***Python, XGBoost, SHAP7*** | **Jan 2026 – Feb 2026**

- Engineered **15+ predictive features** from **66,679** HCRIS hospital-year cost reports (Medicaid share, uncompensated-care burden, occupancy, staffing intensity) and trained an **XGBoost** model with hospital-level grouped cross-validation to forecast operating margins across **6,483 hospitals**, achieving **0.084 MAE** and **0.129 RMSE** on held-out hospitals with **SHAP** surfacing occupancy and Medicaid exposure as dominant drivers.
- Stress-tested every hospital under simulated **5–15% Medicaid reimbursement shocks**, quantifying that low-exposure private urban hospitals (Medicaid <20%) absorbed shocks within **1.2 percentage points** of margin while high-exposure urban teaching hospitals (Medicaid 40–60%) turned financially fragile.
- Scored and ranked all hospitals by a composite resilience metric (Δmargin, post vs. pre-shock), isolating the **top and bottom 20** institutions to give policymakers a data-driven shortlist of the most and least shock-absorptive hospital profiles across **2011–2022**.

Masked Conditioning for Cascaded Diffusion | ***PyTorch, U-Net, Diffusion Models*** **Sep 2025 – Dec 2025**

- Proposed and implemented a channel-masked conditioning variant of the **cascaded diffusion framework** ($64\times64 \to 128\times128$), selectively corrupting **RGB** channels with mask probability 0.35 to improve robustness in **super-resolution** image generation.
- Engineered a two-stage class-conditional diffusion pipeline with **U-Net backbones**, cosine noise schedule over T=1000 timesteps, EMA model averaging (decay=0.995), and **DDIM sampling** (200 steps, η=0), achieving FID $\sim$ 42 at 100 inference steps on a 30-class ImageNet subset ($\sim$33K images).
- Concluded both baseline and masked conditioning converge to comparable fidelity, suggesting the channel-masking structure preserves conditioning quality without degrading the augmentation benefit.

Harmful Brain Activity Classification | ***EfficientNet, TensorFlow, scikit-learn*** **Feb 2025 – May 2025**

- Reduced misclassification of 6 harmful brain activity types by **39%** (KL divergence 0.85 → 0.52) by engineering a **dual-source 8-channel spectrogram pipeline** that fused Kaggle-provided spectrograms with EEG-derived mel spectrograms across **17,089 patient recordings** from the HMS Kaggle competition.
- Accelerated neurologist-grade EEG labeling across **4 brain regions** (LL, RL, LP, RP) by transforming **19 raw electrode-pair signals** into clinically interpretable spectrograms using librosa mel-spectrograms with bipolar montage averaging, log-scaling, and z-score normalization at 200 Hz sampling rate.
- Achieved a best cross-validated KL divergence of **0.52** across **5-fold GroupKFold** (grouped by patient ID to prevent leakage) by training an **EfficientNetB0** classifier on combined 512×512 spectrogram images with cosine LR annealing and mixed-precision inference on dual T4 GPUs.

Auto Insurance Claim Prediction | ***R, XGBoost, Tweedie Regression, Cross-Validation*** **Oct 2024 – Dec 2024**

- Led **R-based modeling and feature selection** for a 60,000-record insurance claim-cost prediction project, handling a highly **semi-continuous dataset** with only **6.8% non-zero claims** and positive costs from $200 to ∼ $60K.
- Compared Logit + Gamma, Tweedie regression, and XGBoost models using an 80/20 train-test split, 5-fold **cross-validation**, over-dispersion checks, Hosmer-Lemeshow testing, and **AIC-based variable selection**.
- Improved final normalized **Gini index to 0.45** and presented **stakeholder-ready risk drivers**, including exposure, vehicle value, max power, term length, vehicle age, and engineered ratio features for **claim-cost planning**.

Large-Scale Twitter Discourse Analysis: Russia-Ukraine Conflict | ***NLP, K-Means*** **Jan 2022 – Apr 2022**

- Engineered an end-to-end Twitter NLP pipeline that scraped **2.9M tweets** over **15 days** and filtered to **943K English posts**, cutting downstream processing noise by ∼**67%**.
- Reduced high-dimensional text feature space across **5 NLP representations** (BoW, n-gram, TF-IDF variants) using **PCA**, retaining **90% variance** while shrinking feature complexity for K-Means clustering.
- Identified **3 dominant discourse themes** in ∼943K war-related tweets using **silhouette- and elbow-optimized K-Means**, surfacing public sentiment patterns across geopolitical, humanitarian, and nuclear threat narratives.

Achievements

- **Won HeatMap Hackathon 2026** (BData Inc. × American Burn Association × MN Healthcare), built a geospatial systems solution translating burn injury data into an equitable care access analysis, selected over competing teams for real-world strategic impact.
- Received **Academic Excellence Award** for outstanding performance in MSc Statistics, IIT Kanpur.